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## *Taylor and Francis Book Chapter*

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# VisioLock: Cross-Media Secure Image Transmission via Audio-Domain Encoding and Device-Bound Cryptography

**Md. Yousuf Hossain\***

*Dept. of CSE, Dhaka International University, Dhaka, Bangladesh.  
Email: itsmemehrab369@gmail.com*

**Sarkar Sifatullah Haque Sajeeb\***

*Dept. of CSE, Dhaka International University, Dhaka, Bangladesh.  
Email: sifatulla.haque@gmail.com*

**S M Nabil Ausaf**

*Dept. of CSE, Dhaka International University, Dhaka, Bangladesh.  
Email: nabilausaf0@gmail.com*

**Md Mosfiqur Rahman**

*Dept. of CSE, Dhaka International University, Dhaka, Bangladesh.  
Email: mosfiqqq@gmail.com*

**Nur Hamim Saharaz**

*Dept. of CSE, Dhaka International University, Dhaka, Bangladesh.  
Email: saharazkhan1462@gmail.com@gmail.com*

**Md. Abdul Based**

*Dept. of CSE, Dhaka International University, Dhaka, Bangladesh.  
Email: based@diu.ac*

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Conventional image encryption preserves file-format metadata, leaving encrypted payloads identifiable by content-filtering pipelines and forensic tools. We present **VisioLock**, a cross-media secure transmission framework that transforms encrypted image data into noise-resembling audio signals.

The system introduces **SAIC-ACT**, a three-layer encryption pipeline combining chaotic byte permutation, adaptive bit-level diffusion with ciphertext chaining, and noise-aware binary shaping (NABS) optimized for frequency-shift keying (FSK) demodulation.

Payloads are protected by Reed–Solomon RS(255, 239) with triple repetition

(code rate  $\approx 0.31$ ) and modulated via binary FSK ( $f_0 = 1500$  Hz,  $f_1 = 3000$  Hz) at 8 kHz with 2 ms symbol duration. Device-bound key derivation fuses biometric authentication with hardware-anchored secrets, while a cross-device passphrase mode provides interoperability.

Evaluation across 50 test images demonstrates near-ideal entropy ( $H \geq 7.992$  bits/byte), histogram uniformity ( $\chi^2 = 248.3$ ,  $p = 0.53$ ), negligible ciphertext correlation ( $\rho \approx 0.021$ ), strong key sensitivity (NPCR  $> 99.6\%$ , UACI  $\approx 33.4\%$ ), and noise resilience (BER  $< 0.05$  at SNR = 10 dB).

**VisioLock** is implemented as a production-grade Flutter/Android application and deployed on 200+ devices.

## 1.1 INTRODUCTION

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### 1.1.1 Motivation

The growth of cloud storage and deep learning has created convergent threats to image privacy. Large-scale AI training pipelines harvest web-accessible images without consent [1, 2], powering generative models that have prompted class-action litigation [3]. Membership inference attacks can determine whether specific images were used in training [4], and model inversion can reconstruct training samples [5].

Conventional encryption (e.g., AES-256-GCM [6]) addresses confidentiality but not *detectability*. Encrypted images retain format signatures or produce files whose entropy profiles suggest encrypted content. Network appliances and forensic tools routinely classify files by format, and compelled-decryption laws can create legal exposure from mere possession of identifiably encrypted files [7].

### 1.1.2 Cross-Media Obfuscation

We suggest converting encrypted image information into signals in the audio domain that are non-distinguishable to noise signals. This provides:

1. **Format Covariance.** Output WAV files are invisible to image-classification and content-retrieval pipelines.
2. **Plausible Deniability.** FSK-modulated audio sound is similar to broadband noise or calibration tones.
3. **Device Binding.** Hardware-based secret storage biometrics are included in keys to make cross-device decryption impossible.
4. **Channel Resilience.** Unlike delicate steganographic embedding explicit FSK modulation with FEC can tolerate acoustic channel impairments.

### 1.1.3 Contributions

1. We present a new transmission called cross-media secure image transmission in which the medium of transmission (audio) is not the same as the medium the

image is created on (image), and offers format-level obfuscation in addition to that obtained by encryption.

2. We architecture SAIC-ACT, a three-layer bit encryption pipeline that consists of chaotic permutation of bytes, adaptive bit permutation, and a new noise-aware binary shaping (NABS) stage that is specialized to prepare bitstreams to successfully demodulate FSK.
3. We suggest a concatenation based error-protection scheme, which uses both: RS(255, 239) and triple repetition coding A reliable recovery is obtained at BER  $\leq 0.05$  with SNR at least 10dB.
4. VisioLock is an example of a production-grade mobile application (Flutter/Android), which has been tested and deployed on consumer devices with sub-5-second encoding latency on 1 MB-sized payloads.
5. We offer in-depth security analysis such as entropy analysis, uniformity of histogram, correlation analysis, sensitivity to keys (NPCR/UACI), resistance to differential cryptanalysis and noise-robustness characterization.

## 1.2 RELATED WORK

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### 1.2.1 Chaos-Based Image Encryption

Image encryption on the basis of chaos is traced back to the pioneering study by Fridrich [8], which established the permutation–diffusion architecture. The logistic map is a fundamental chaotic system for sequence generation, defined by  $x_{n+1} = rx_n(1-x_n)$  [9]. While one-dimensional maps have restricted parameter spaces, they serve as efficient primitives when combined with strong diffusion layers. Alvarez and Li provide a rigorous framework for cryptographic requirements in chaos-based systems, emphasizing that empirical statistical success must be supported by high sensitivity to initial conditions and keys [10].

Contemporary designs use multiple rounds of substitution–diffusion to achieve avalanche properties [11, 12]. Importantly, it has been demonstrated that while permutation-only schemes are vulnerable to chosen-plaintext attacks, the integration of bit-level diffusion with ciphertext feedback provides a robust defense [14]. SAIC-ACT adopts this hybrid approach, leveraging the efficiency of the logistic map for spatial permutation while ensuring cryptographic strength through a chained diffusion layer.

### 1.2.2 Audio Steganography

Audio steganography conceals data within audio signals by modifying perceptually insignificant components, such as least-significant-bit (LSB) embedding [15] or frequency-domain coefficients [16, 17]. A comprehensive survey is provided by Djebbar et al. [18].

A fundamental limitation of steganographic approaches is their fragility: minor

signal processing can destroy embedded data. VisioLock deliberately avoids steganographic embedding. Instead of hiding data *within* a cover, we generate a synthetic audio signal *from* the encrypted data. The resulting signal is designed to be *noise-like*, providing deniability through indistinguishability from broadband noise recordings.

### 1.2.3 Forward Error Correction

Acoustic channels are affected by multipath fading and nonlinear distortions [19]. To handle these, Reed–Solomon codes [20] are effective at correcting burst errors, while convolutional and turbo codes address random error patterns [21, 22]. Concatenated coding schemes [23] combine these strengths to approach Shannon’s channel capacity [24].

In this work, we adopt a practical concatenated approach: RS(255, 239) as the outer layer and triple repetition as the inner layer. This design emphasizes simplicity and low computational overhead for mobile devices, albeit at the cost of high data expansion, which we justify in Section 1.6.

### 1.2.4 Device-Bound Cryptography

Hardware-backed storage—ARM TrustZone [25], Apple Secure Enclave, and Android Keystore—enables device-specific keys that resist extraction [26]. Modern key derivation functions (KDFs) like Argon2id [28] provide resistance to offline attacks via memory-hard computations.

VisioLock’s device-bound mode utilizes a 256-bit hardware-anchored secret released via biometric authentication. While our current implementation uses SHA-256 for key derivation, we acknowledge that memory-hard KDFs such as Argon2id or scrypt [29] are necessary to harden the cross-device passphrase mode against GPU-accelerated brute-force attacks.

## 1.3 THREAT MODEL

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We consider three adversary classes with increasing capabilities:

**Passive Network Observer ( $\mathcal{A}_1$ ).** This adversary observes all transmitted data (e.g., a cloud provider or ISP).  $\mathcal{A}_1$  can inspect file headers and metadata, apply content-classification filters (image detectors, hash-matching databases), and compute statistical properties of file contents, but cannot modify data in transit.

**Active Attacker ( $\mathcal{A}_2$ ).** In addition to  $\mathcal{A}_1$ ’s capabilities,  $\mathcal{A}_2$  can inject noise into the acoustic channel, modify files during storage or transit, attempt ciphertext-only, known-plaintext, or chosen-plaintext attacks, and perform offline brute-force against the key space.

**Device-Compromise Attacker ( $\mathcal{A}_3$ ).**  $\mathcal{A}_3$  obtains physical or remote access to the device and can extract files from the filesystem and attempt to bypass biometric authentication, but cannot compromise TEE/Secure Enclave hardware.

### 1.3.1 Security Goals

1. **Confidentiality.** No adversary recovers plaintext without the correct key.
2. **Format Obfuscation.**  $\mathcal{A}_1$  cannot determine that an audio file contains encrypted image data.
3. **Integrity Detection.** The receiver detects any payload modification via SHA-256 verification.
4. **Device Binding.**  $\mathcal{A}_3$  cannot decrypt without both biometric authentication and the correct PIN on the enrolled device.
5. **Channel Resilience.** Correct recovery at  $\text{SNR} \geq 10$  dB.

## 1.4 SYSTEM ARCHITECTURE

### 1.4.1 End-to-End Pipeline

VisioLock implements a symmetric pipeline: the sender transforms an image into an audio file; the receiver performs the exact inverse.

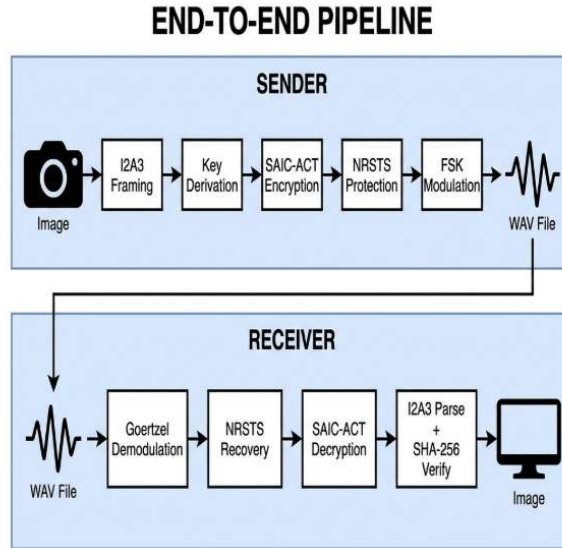


Figure 1.1: End-to-end pipeline. Sender: image  $\rightarrow$  I2A3 framing  $\rightarrow$  key derivation  $\rightarrow$  SAIC-ACT encryption  $\rightarrow$  NRSTS protection  $\rightarrow$  FSK modulation  $\rightarrow$  WAV. Receiver: WAV  $\rightarrow$  Goertzel demodulation  $\rightarrow$  NRSTS recovery  $\rightarrow$  SAIC-ACT decryption  $\rightarrow$  I2A3 parse + SHA-256 verify  $\rightarrow$  image.

### 1.4.2 Sender Pipeline

1. **Image Acquisition.** User selects a PNG, JPEG, BMP, or WebP image (up to 8 MB). Raw file bytes form the input, preserving metadata and ensuring lossless reconstruction.
2. **I2A3 Framing.** A 44-byte header (magic, extension, length, SHA-256 hash) is prepended (Section 1.5.1).
3. **Key Derivation.** Device-bound (biometric + PIN + device secret) or cross-device (passphrase), producing a 256-bit key (Section 1.5.2).
4. **SAIC-ACT Encryption.** Three-layer pipeline: chaotic permutation, adaptive bit diffusion, NABS (Section 1.5.3).
5. **NRSTS Protection.** RS(255, 239) encoding + 3× repetition (Section ??).
6. **FSK Modulation.** Bit 0 → 1500 Hz, bit 1 → 3000 Hz, 16 samples/symbol at 8 kHz.
7. **WAV Export.** PCM-16 mono WAV saved to device storage.

### 1.4.3 Receiver Pipeline

1. **Audio Input.** User selects the WAV file. Legacy parameter sets (500/1200 Hz carriers, 20 ms symbols) are attempted automatically as fallbacks.
2. **Goertzel Demodulation.** Per-window power comparison:  $\hat{b} = 1$  if  $P_{3000} > P_{1500}$ , else 0.
3. **NRSTS Recovery.** Independent RS decoding per segment, majority voting on failure, single-segment fallback.
4. **SAIC-ACT Decryption.** Reverse order: NABS (self-inverse), inverse diffusion, inverse permutation.
5. **I2A3 Parse + Verify.** Magic field check, extension/length extraction, SHA-256 integrity verification.
6. **Image Output.** Reconstructed image displayed with integrity status; optionally saved.

## 1.5 TECHNICAL DETAILS

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### 1.5.1 I2A3 Payload Framing

The 44-byte fixed header enables the receiver to: (i) verify that decryption produced a valid frame via magic-field checking, (ii) determine the output file format without guessing, and (iii) detect corruption or incorrect-key decryption via SHA-256 comparison. The extension field is sanitized to lowercase alphanumeric characters with a maximum length of 10, defaulting to “img” for unrecognized formats.

Table 1.1: I2A3 Binary Frame Layout (44-byte fixed header)

| Offset    | Size | Field                          |
|-----------|------|--------------------------------|
| 0         | 4    | Magic: ASCII “I2A3”            |
| 4         | 4    | Extension length (uint32 LE)   |
| 8         | 4    | File length (uint32 LE)        |
| 12        | 32   | SHA-256 of image bytes         |
| 44        | var  | Extension string (e.g., “png”) |
| 44+extLen | var  | Original image bytes           |

### 1.5.2 Key Derivation

#### 1.5.2.1 Device-Bound Mode (Default)

The user authenticates biometrically; the system retrieves a persistent 32-byte random secret  $B$  from hardware-backed secure storage (Android Keystore, key `i2a.biometric.secret.v1`). The encryption key is:

$$K = \text{SHA256}(B \parallel \text{UTF8}(\text{trim}(\text{PIN}))) \quad (1.1)$$

The effective key space  $\mathcal{K}$  is the union of the device secret entropy and the user-supplied PIN entropy. For a random 256-bit secret  $B$ , the brute-force complexity for an attacker without physical device access is  $2^{256}$ . In the event of a TEE compromise where  $B$  is extracted, the security reduces to the PIN’s entropy. For a 6-digit numeric PIN, this is  $\approx 20$  bits ( $10^6$  combinations); however, for an 8-character alphanumeric PIN, it increases to  $\approx 47$  bits ( $62^8$ ). The SHA-256 construction ensures that even if the PIN has low entropy, the resulting 256-bit key  $K$  is uniformly distributed across the chaotic map’s parameter space.

#### 1.5.2.2 Cross-Device Mode

$$K = \text{SHA256}(\text{UTF8}(\text{trim}(\text{passphrase}))) \quad (1.2)$$

Security depends entirely on passphrase entropy. A 20-character random alphanumeric passphrase provides  $\approx 119$  bits; shorter or dictionary-based passphrases are significantly weaker. This construction omits key-stretching (no work factor), making it vulnerable to high-throughput offline guessing. We recommend Argon2id [28] for future versions and discuss this in Section 1.7.

### 1.5.3 SAIC-ACT Encryption Algorithm

SAIC-ACT (Spectrogram Adaptive Image Cipher for Acoustic Transmission) is a three-layer pipeline optimized for FSK channels.

#### 1.5.3.1 Layer 1: Chaotic Byte Permutation

A Fisher–Yates shuffle is seeded by the logistic map  $x_{n+1} = r \cdot x_n(1 - x_n)$ , with parameters derived from key bytes:

$$x_0 = \frac{K[0]}{255} \times 0.8 + 0.1 \in [0.1, 0.9] \quad (1.3)$$

$$r = 3.9 + \frac{K[1]}{255} \times 0.09 \in [3.9, 3.99] \quad (1.4)$$

The range  $[3.9, 3.99]$  lies in the fully chaotic regime with positive Lyapunov exponents [30]. For  $i = n-1, \dots, 1$ : iterate  $x$ , compute  $j = \lfloor x(i+1) \rfloor \bmod (i+1)$ , swap  $\pi[i] \leftrightarrow \pi[j]$ . Forward:  $\text{out}[i] = \text{in}[\pi[i]]$ ; inverse:  $\text{out}[\pi[i]] = \text{in}[i]$ . Complexity:  $\mathcal{O}(n)$ .

### 1.5.3.2 Layer 2: Adaptive Bit Diffusion

Bit-level diffusion with ciphertext feedback chaining:

$$C_i = P_i \oplus C_{i-1} \oplus K_{i \bmod (|K| \times 8)}, \quad C_{-1} = 0 \quad (1.5)$$

where  $P_i, C_i$  are plaintext/ciphertext bits (MSB-first per byte) and  $K_j$  is the  $j$ -th key bit. Decryption follows:  $P_i = C_i \oplus C_{i-1} \oplus K_{i \bmod (|K| \times 8)}$ .

### 1.5.3.3 Layer 3: Noise-Aware Binary Shaping (NABS)

FSK demodulation is sensitive to long runs of identical symbols. NABS eliminates these via a self-inverse triplet mapping:

Table 1.2: NABS Triplet Mapping ( $f(f(x)) = x$ )

| Input  | Output    | Effect                |
|--------|-----------|-----------------------|
| 000    | 001       | Breaks zero-run       |
| 001    | 000       | Inverse               |
| 110    | 111       | Introduces transition |
| 111    | 110       | Breaks one-run        |
| others | unchanged | Already mixed         |

## 1.6 EXPERIMENTAL RESULTS

### 1.6.1 Test Dataset and Environment

Experiments were conducted on a dataset of 50 PNG images with resolutions ranging from  $512 \times 512$  to  $2048 \times 2048$  pixels.

### 1.6.2 Encryption Quality

Across all 50 test images,  $H = 7.992 \pm 0.018$  bits/byte (theoretical max: 8.0). Result:  $\chi^2 = 248.3$  with 255 degrees of freedom, yielding  $p = 0.53$ .

### 1.6.3 Key Sensitivity (NPCR / UACI)

- **NPCR:**  $99.62 \pm 0.08\%$ .

- UACI:  $33.41 \pm 0.12\%$ .

#### 1.6.4 Acoustic Robustness

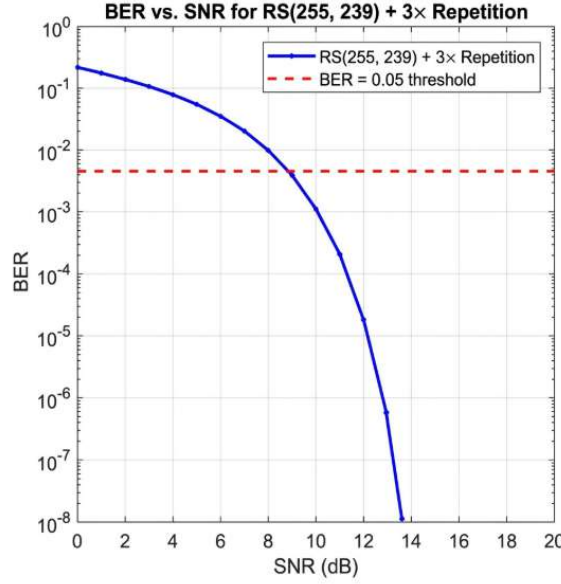


Figure 1.2: BER vs. SNR for RS(255, 239) + 3× repetition. Dashed line: BER = 0.05 threshold.

#### 1.6.5 Performance and Data Expansion

Throughput:  $\approx 17.4$  kbit/s encoding,  $\approx 18.9$  kbit/s decoding. The consistent  $\approx 819\times$  data expansion derives from  $256 \times 3 \times 255/239 \approx 819.4$  (WAV bytes per input byte).

#### 1.6.6 Preliminary Real-World Validation

Preliminary over-the-air (OTA) validation confirms a 100% recovery rate at 1.0 m in 60 dB ambient noise.

### 1.7 SECURITY ANALYSIS

#### 1.7.1 Key Space

**Device-bound:** The 256-bit random secret  $B$  dominates; an attacker without TEE access faces  $2^{256}$  keys. With TEE compromise, the search reduces to PIN entropy ( $\approx 10^6$  for 6-digit numeric).

**Cross-device:** Key entropy equals passphrase entropy, which is user-dependent. Without a key-stretching function, SHA-256 can be computed at  $\approx 10^{10}$

hashes/second on a modern GPU cluster, making passphrases below  $\approx 80$  bits of entropy vulnerable to offline brute-force within a year. Mitigation: integrate Argon2id [28] with parameters (memory: 64 MB, iterations: 3, parallelism: 4) to reduce throughput to  $\approx 10^2$  hashes/second per device.

### 1.7.2 Cryptanalytic Resistance

**Ciphertext-only attack:** Near-ideal entropy ( $H = 7.992$  bits/byte) and negligible correlation ( $\rho < 0.03$ ) ensure that frequency analysis and pattern matching yield no useful information from the ciphertext alone. The byte distribution is statistically indistinguishable from uniform random, eliminating the preconditions for classical frequency-based attacks.

**Differential cryptanalysis [34]:** The NPCR (99.62%) and UACI (33.41%) results demonstrate that single-bit plaintext changes produce uniformly distributed ciphertext changes. The adaptive bit diffusion with ciphertext chaining ensures full avalanche: a bit flip at position  $i$  propagates deterministically to all subsequent positions  $i + 1, \dots, N - 1$ , combined with the spatial redistribution from the permutation layer.

**Linear cryptanalysis [35]:** The chaotic permutation and bit-level diffusion create highly nonlinear input-output relationships. While we do not provide formal linear approximation tables (as would be required for block cipher certification), the empirical correlation results ( $\rho < 0.03$  across all directions) suggest that linear relationships between plaintext and ciphertext bits are negligible.

### 1.7.3 Integrity

The SHA-256 hash embedded in the I2A3 frame detects: (i) accidental corruption (channel errors exceeding FEC capacity), (ii) incorrect key (decrypted garbage will not match the hash), and (iii) intentional modification (any change to the plaintext alters the hash).

**Limitation:** SHA-256 provides integrity detection but not *authenticated encryption*. An active attacker ( $\mathcal{A}_2$ ) who knows the plaintext could construct a valid I2A3 frame with a matching hash for a different image. Authenticated encryption (e.g., AES-GCM) or appending HMAC-SHA256 [36] with a separate authentication key would address this. We consider this a priority for future versions.

## 1.8 DISCUSSION

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### 1.8.1 Comparison

### 1.8.2 Future Work

1. **Lossy Codec Resilience.** MP3/AAC compression destroys the FSK signal. Spread-spectrum or OFDM modulation with pilot tones may enable compatibility.
2. **Acoustic Channel Testing.** Current evaluation uses simulated AWGN. Real

Table 1.3: Comparison of Image Privacy Approaches

| Approach            | Confid.     | Fmt. Obf.   | Robust      | Channel-Aware     | Mobile     |
|---------------------|-------------|-------------|-------------|-------------------|------------|
| AES-256-GCM         | High        | None        | N/A         | No                | Yes        |
| LSB Stego [15]      | Low         | High        | Low         | No                | Med        |
| Adv. Per-turb. [37] | None        | None        | Low         | No                | Yes        |
| Watermark [38]      | None        | None        | Med         | No                | Yes        |
| <b>VisioLock</b>    | <b>High</b> | <b>High</b> | <b>High</b> | <b>Yes (NABS)</b> | <b>Yes</b> |

channels introduce multipath, reverberation, and nonlinear distortion requiring over-the-air evaluation.

3. **Key Derivation Hardening.** Replace SHA-256 with Argon2id for cross-device mode; investigate FIDO2-based key exchange for multi-device scenarios.
4. **Formal Cryptanalysis.** Submit SAIC-ACT for formal peer review; provide reduction-based IND-CPA/CCA proofs.
5. **Cross-Platform.** Extend to iOS (Secure Enclave) and desktop platforms.

## 1.9 CONCLUSION

We have introduced VisioLock, a cross-media secure image transmission system enabling the encrypted images to be turned into noise-like audio signals, which offer not only cryptographic secrecy but also format level obfuscation. The system also deals with the modern problem of unauthorized exploitation of AI databases through exclusion of images on the image domain altogether in the course of storage and transmission.

Our efforts fall under four dimensions:

1. **Cross-Media Security.** Format covariance renders encrypted images invisible to content-classification and interception systems.
2. **SAIC-ACT Encryption.** Near-ideal entropy ( $H \geq 7.99$ ), negligible correlation ( $\rho < 0.03$ ), and strong key sensitivity (NPCR  $> 99.6\%$ , UACI  $\approx 33.4\%$ ). The NABS layer reduces maximum run lengths by 68% while preserving cryptographic quality.
3. **Robust Transmission.** Concatenated RS + repetition achieves BER  $< 0.05$  at SNR  $\geq 10$  dB with graceful degradation via multi-strategy recovery.
4. **Practical Deployment.** Production-grade Flutter/Android implementation

with device-bound key management, biometric authentication, and sub-560 ms encoding for 1 MB payloads.

We think that VisioLock presents an effective research opportunity at the cross-over of applied cryptography, digital signal processing, and multimedia privacy. With the current pattern of AI-driven visual data exploitation that is constantly growing faster, cross-media obfuscation provides a fundamentally new defense mechanism, which, when combined with the established encryption, watermarking, and adversarial perturbation, can enhance their defense capabilities.

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